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Batch: CSE(DS)-57

Assignment 1:

```
name = input("Enter student name: ")
mark1 = int(input("Enter Mark 1: "))
mark2 = int(input("Enter Mark 2: "))
mark3 = int(input("Enter Mark 3: "))

average = mark1 + mark2 + mark3 / 3

if average >= 50:
    print(name, "Pass")
else:
    print(name, "Fail")
```

Ans) Problem 1 The expression `mark1+mark2+mark3/3` divides only `mark3` by 3 due to division precedence before addition. It should use brackets: `(mark1+mark2+mark3)/3`.

Problem 2 The average of 3 numbers is usually sound the range of 40 to 50 but in the code it is given greater than 50. This may result in fail for most inputs.

Problem 3 No Validation of marks, the program can accept any integer as input even negative numbers.

AI suggestions accepted/rejected

Accepted:

- Correcting the average calculation.
- Fixing Python indentation.
- Adding validation for marks between 0 and 100.
- Using `try-except` to handle non-numeric input.
- Calculating the average using `sum()`.

Rejected/modified:

- Making the code too complicated like using unnecessary functions and making the program too long

Correct code given by AI:

```
error.py 2 ...
1 def get_mark(mark_number):
2     while True:
3         try:
4             mark = int(input(f"Enter Mark {mark_number} (0-100): "))
5         except ValueError:
6             print("Please enter a whole number.")
7             continue
8
9         if 0 <= mark <= 100:
10            return mark
11
12    print("Mark must be between 0 and 100.")
13
14
15 name = input("Enter student name: ")
16 mark1 = get_mark(1)
17 mark2 = get_mark(2)
18 mark3 = get_mark(3)
19
20 average = (mark1 + mark2 + mark3) / 3
21
22 if average >= 50:
23     print(name, "Pass")
24 else:
25     print(name, "Fail")
26
```

Identify three errors

correct the code. add validation for 0-100...

The edit is in place. I'm validating both compilation and the retry behavior with invalid mark followed by valid input, so the check covers the requested cases rather than only syntax.

Executed command with input for error.py

1 confirmation pending

Run pwsh command?

python -m py_compile .\error.py

Compiles error.py to verify Python syntax.

```
KeyboardInterrupt
PS C:\Users\Afifa\OneDrive\Documents\GitHub\AI assisted coding> C:\Users\Afifa\AppData\Local\Microsoft\Windows\Terminal>
e "c:/Users/Afifa/OneDrive/Documents/GitHub/AI assisted coding/error.py"
Enter student name: abcd
Enter Mark 1 (0-100): 50
Enter Mark 2 (0-100): 60
Enter Mark 3 (0-100): 90
abcd Pass
PS C:\Users\Afifa\OneDrive\Documents\GitHub\AI assisted coding> & C:\Users\Afifa\AppData\Local\Microsoft\Windows\Terminal>
exe "c:/Users/Afifa/OneDrive/Documents/GitHub/AI assisted coding/error.py"
Enter student name: Rishi
Enter Mark 1 (0-100): 70
Enter Mark 2 (0-100): 120
Mark must be between 0 and 100.
Enter Mark 2 (0-100): 70
Enter Mark 3 (0-100): 50
Rishi Pass
PS C:\Users\Afifa\OneDrive\Documents\GitHub\AI assisted coding> & C:\Users\Afifa\AppData\Local\Microsoft\Windows\Terminal>
exe "c:/Users/Afifa/OneDrive/Documents/GitHub/AI assisted coding/error.py"
Enter student name: Tina
Enter Mark 1 (0-100): abd
Please enter a whole number.
Enter Mark 1 (0-100): 67
Enter Mark 2 (0-100): 34
Enter Mark 3 (0-100): 45
Tina Fail
PS C:\Users\Afifa\OneDrive\Documents\GitHub\AI assisted coding>
```

Assignment 2

```
n = int(input("Enter n: "))

total = 0

for i in range(1, n):
    if i % 2 == 1:
        total = total + i

print("Sum of even numbers =", total)
```

Ans) The logical error in this program are:

- 1) `if%2==1`: it calculates for odd numbers and not for even numbers
- 2) `for i in range(1,n)`: the loop will terminate at `n`, it will not include the number `n`. Example if `n=10` the loop will take the numbers from 1 to 9

The output is wrong because of the statement `if%2==1`

Correct code given by AI:

```
1  n = int(input("Enter n: "))
2
3  total = 0
4
5  for i in range(1, n + 1):
6      if i % 2 == 0:
7          total = total + i
8
9  print("Sum of even numbers =", total)
10
```

n	Even Number	Manual Calculation	Output
10	2,4,6,8,10	2+4+6+8+10=30	30
20	2,4,6,8,10,12,14,16,18,20	2+4+6+8+10+12+14+16+18+20=110	110
1	No even numbers	0	0

Short debugging explanation

The original program produced the wrong result because it checked for odd numbers using `i % 2 == 1` instead of even numbers using `i % 2 == 0`. The loop also excluded `n` because `range(1, n)` stops at `n-1`. I corrected the condition and changed the loop to `range(1, n + 1)`. The corrected program gives 30 for `n=10`, 110 for `n=20`, and 0 for `n=1`, which matches the manual calculations.

Assignment 3

```
def find_largest(a, b, c):
    if a > b:
        largest = a
    elif b > c:
        largest = b
    else:
        largest = c

def display_result(x):
    print("Largest number =", x)

x = int(input("Enter first number: "))
y = int(input("Enter second number: "))
z = int(input("Enter third number: "))

result = find_largest(x, y, z)
display_result(result)
```

Ans)The function `find_largest()` calculates the largest number but does not return the value to the calling program.

Because there is no return statement, the variable `result` receives `None` instead of the largest number.

KeyboardInterrupt

```
PS C:\Users\Afifa\OneDrive\Documents\GitHub\AI assisted coding> C:\Users\Afifa\AppData\Local\Python\bin\python.exe "c:/Users/Afifa/OneDrive/Documents/GitHub/AI assisted coding/error.py"
Enter first number: 10
Enter second number: 20
Enter third number: 30
Largest number = 30
PS C:\Users\Afifa\OneDrive\Documents\GitHub\AI assisted coding> & C:\Users\Afifa\AppData\Local\Python\bin\python.exe "c:/Users/Afifa/OneDrive/Documents/GitHub/AI assisted coding/error.py"
Enter first number: 50
Enter second number: 20
Enter third number: 10
Largest number = 50
PS C:\Users\Afifa\OneDrive\Documents\GitHub\AI assisted coding> & C:\Users\Afifa\AppData\Local\Python\bin\python.exe "c:/Users/Afifa/OneDrive/Documents/GitHub/AI assisted coding/error.py"
Enter first number: 20
Enter second number: 50
Enter third number: 30
Largest number = 50
PS C:\Users\Afifa\OneDrive\Documents\GitHub\AI assisted coding> & C:\Users\Afifa\AppData\Local\Python\bin\python.exe "c:/Users/Afifa/OneDrive/Documents/GitHub/AI assisted coding/error.py"
Enter first number: 25
Enter second number: 25
Enter third number: 10
Largest number = 25
PS C:\Users\Afifa\OneDrive\Documents\GitHub\AI assisted coding> █
```